charge. However, later experiments showed that under certain circumstances, they exhibit wave-like properties also.

In this unit, the particle nature of waves (radiation) and the wave nature of particles (matter) – that is, wave-particle duality of radiation and matter is discussed with the relevant experimental observations supporting this dual nature.

### 8.1.1 Electron emission

In metals, the electrons in the outer most shells are loosely bound to the nucleus. Even at room temperature, there are a large number of free electrons which are moving inside the metal in a random manner. Though they move freely inside the metal, they cannot leave the surface of the metal. The reason is that when free electrons reach the surface of the metal, they are attracted by the positive nuclei of the metal. It is this attractive pull which will not allow free electrons to leave the metallic surface at room temperature.

In order to leave the metallic surface, the free electrons must cross a potential barrier created by the positive nuclei of the metal. **The potential barrier which prevents free electrons from leaving the metallic surface is called surface barrier**.

Free electrons possess some kinetic energy and this energy is different for different electrons. The kinetic energy of the free electrons is not sufficient to overcome the surface barrier. Whenever an additional energy is given to the free electrons, they will have sufficient energy to cross the surface barrier and they escape from the metallic surface**. The liberation of electrons from any surface of a substance is called electron emission**.

**The minimum energy needed for an electron to escape from the metal surface is called work function of that metal**. The work function of the metal is denoted by $\phi_0$ and is measured in electron volt (eV).

**Note**

The SI unit of energy is joule. But electron volt is a commonly used unit of energy in atomic and nuclear physics.

One electron volt is defined as the kinetic energy gained by an electron when accelerated by a potential difference of 1 V.

1 *eV* = KE gained by the electron

= Work done by the electric field

= q V

= $1.602 \times 10^{-19}$ C × 1 V

= $1.602 \times 10^{-19}$ J

Suppose the maximum kinetic energy of the free electron inside the metal is 0.5 eV and the energy needed to overcome the surface barrier of a metal is 3 eV, then the minimum energy needed for electron emission from the metallic surface is 3 – 0.5 = 2.5 eV. Here 2.5 eV is the work function of the metal.

The work function is different for different metals and is a typical property of metals and the nature of their surface. Table 8.1 gives the approximate value of work function for various metals. The material with smaller work function is more effective in electron emission because extra energy required to release the free electrons from the metal surface is smaller.



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